OurMocks

2027 CYCLE

MOCK · PRACTICE · CRACK

CUET UG

Mathematics

Time Based Data

Time Series, Components and Trend Measurement · Section B2 V · CUET UG 2027 Cycle Notes

INSIDE
Concept Notes

INSIDE
Solved Examples

INSIDE
Practice MCQs + Solutions

Aligned to the latest official syllabus · Concept-first explanations

www.ourmocks.com

Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) 2026 syllabus — Mathematics/Applied Mathematics (319), the current 2027 baseline)	Skills the questions test
Time series and univariate analysis	"Time Series: Identify time series as chronological data. Time Series analysis for univariate data: Solve practical problems based on statistical data and Interpret." (Section B2 Applied Mathematics Unit V)	Recognising chronological structure; reading and interpreting a series.
Components of a time series	"Components of Time Series: Distinguish between different components of time series." (Section B2 Applied Mathematics Unit V)	Naming trend / seasonal / cyclical / irregular from a described series.
Secular trend	"Secular trend: Understand the long term tendency." (Section B2 Applied Mathematics Unit V)	Separating the long-run direction from short-term noise.
Methods of measuring trend	"Methods of Measuring trend: Demonstrate the techniques of finding trend by different methods." (Section B2 Applied Mathematics Unit V)	Freehand, semi-averages, moving averages and least squares, each with its care points.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The 2026 syllabus houses this unit as **Section B2 Applied Mathematics Unit V** — the data-over-time toolkit, where most marks come from **classifying** a movement correctly before any computation begins.

What a Time Series Is — Data With a Clock

A **time series** is a single variable recorded at successive times — yearly rainfall, monthly sales, daily temperature — and the clock is not decoration: the **order** carries the meaning. Shuffle the entries and the series stops being a series; the whole subject lives on that ordering. It is **univariate** data: one measured quantity paired with its time stamp, nothing else. Reading a series fluently — spotting its direction, its rhythm, its shocks — is the interpretation skill the syllabus names, and everything else in this chapter is machinery for doing it precisely.

- **Equal spacing is the norm.** Yearly, quarterly or monthly intervals at fixed gaps keep comparisons honest; an irregular grid distorts every trend line drawn on it.
- **Two families of questions follow.** Classification questions — "which component is this movement?" — and computation questions — "find the trend by method X". Both are trained below.

The Four Components — T, S, C, I

Every movement in a series is a blend of four voices. **Trend** (T) is the long-run direction — growth, decline or plateau over years, driven by slow forces like population and technology. **Seasonal** (S) is the calendar-fixed rhythm repeating within a year: ice-cream every summer, woollens every winter, festival sales every October–November. **Cyclical** (C) is the multi-year wave of booms and recessions — like seasonal in repeating, but **not** tied to a calendar and typically spanning several years. **Irregular** (I) is the residue: strikes, floods, viral surprises — one-off shocks with no pattern and no warning.

Component	Signature	Quick example
Trend (T)	Long-run, slow, persistent direction	Smartphone users rising for a decade
Seasonal (S)	Repeats on a fixed calendar, within a year	Umbrella sales every monsoon
Cyclical (C)	Multi-year waves, no fixed calendar	Boom-recession swings in construction
Irregular (I)	One-off, unpredictable spikes or dips	A strike halting a factory for a week

OurMocks

The components combine in two standard models: **additive**, $Y = T + S + C + I$ (absolute swings of similar size), and **multiplicative**, $Y = T \cdot S \cdot C \cdot I$ (swings that grow with the level of the series). The classification habit — calendar-fixed? seasonal; multi-year? cyclical; one-off? irregular; decade-long? trend — answers a full block of MCQs on its own.

WORKED EXAMPLE — NAME THAT MOVEMENT

An ice-cream brand's monthly sales rise over the years, peak every May–June, dip during a two-month supply strike, and ride a three-year boom in the local economy. Classify each movement.

Solution. The years-long rise: **trend**. The May–June peak every single year: **seasonal** — fixed calendar, within a year. The two-month strike dip: **irregular** — one-off shock. The three-year boom: **cyclical** — a multi-year wave with no fixed month attached. Four sentences, four components; this is the most repeated question shape of the unit.

Secular Trend — the Long-Run Direction

Secular trend is the syllabus's name for the long-term tendency — the slow spine of the series that survives when seasons pass, cycles turn and shocks fade. A decade of rising literacy, a steady fall in camera prices, a flat rural electrification rate: each is a trend statement about **decades**, not months. The danger is impatience: a single festive spike or a bad quarter is **not** trend — it is seasonal or irregular noise wearing a trend costume. Measuring the trend means extracting that spine with a repeatable method, and the syllabus names four.

Measuring Trend — Four Methods, Four Care Points

Method 1 — Freehand (graphical). Plot the points; draw a smooth line through their middle by eye. Instant and honest about its subjectivity — two analysts draw two lines, and no two runs agree exactly. Excellent for a first look; weak as a reproducible answer.

Method 2 — Semi-averages. Split the series into **two equal halves**, average each half, plot the two means at the midpoints of their halves, and join — that straight line is the trend. The care point: the halves must be **equal**, so an odd number of years forces you to drop the middle one. Worked: six years with values 10, 12, 14, 16, 18, 20 — first-half mean $\frac{10+12+14}{3} = 12$ at year 2, second-half mean $\frac{16+18+20}{3} = 18$ at year 5; the trend line rises 6 over 3 years, so 2 per year.

Method 3 — Moving averages. Average each span of 3 (or 4) consecutive values and park the average at the span's middle year; slide the window one step along. The wiggles — seasonal and irregular — cancel inside the averages, leaving a smooth ribbon that **is** the trend estimate. A 3-year MA lands on years; a 4-year MA lands **between** two years and needs a second 2-point centring pass. The price: the first and last spans have no average — the ends of the table go dark.

OurMocks

WORKED EXAMPLE — A 3-YEAR MOVING AVERAGE TABLE

Sales: 2019: 12, 2020: 15, 2021: 14, 2022: 18, 2023: 17. Compute the 3-year moving averages.

Solution. Slide the window, park at the middle year:

Year	Sales	3-year moving average
2019	12	—
2020	15	$\frac{12+15+14}{3} = 13.67$
2021	14	$\frac{15+14+18}{3} = 15.67$
2022	18	$\frac{14+18+17}{3} = 16.33$
2023	17	—

The MA column climbs $13.67 \to 15.67 \to 16.33$ — a clean rising trend read off a bumpy series. The two dashes are the method's signature cost.

Method 4 — Least squares (the fitted line). Fit $y = a + bx$ so that the **sum of squared vertical misses** is smallest. With the years **coded** so that $\Sigma x = 0$ (odd count: $-2, -1, 0, 1, 2$; even count: half-steps like $-2.5, -1.5, \ldots$), the formulas collapse to two divisions:

$$a = \frac{\Sigma y}{n}, \quad b = \frac{\Sigma xy}{\Sigma x^2}$$

— a is the trend value at the central year, b the yearly growth the line asserts.

WORKED EXAMPLE — LEAST SQUARES FROM SCRATCH

Fit a straight-line trend to: 2019: 10, 2020: 12, 2021: 13, 2022: 11, 2023: 15.

Solution. Code the years $x = -2, -1, 0, 1, 2$ (so $\Sigma x = 0$). Table:

$$\Sigma y = 10 + 12 + 13 + 11 + 15 = 61, \quad \Sigma xy = (-2)(10) + (-1)(12) + 0 + (1)(11) + (2)(15) = 9, \quad \Sigma x^2 = 10.$$

Then $a = \frac{61}{5} = 12.2$ and $b = \frac{9}{10} = 0.9$. Trend line:

$$y = 12.2 + 0.9x.$$

Read it: the central year 2021 carries the trend value 12.2, and the line claims sales grow 0.9 units per year. Forecast for 2024 ($x = 3$): $12.2 + 2.7 = 14.9$ — one substitution, the method's payoff.

EVERYDAY EXAMPLE — THE NOISY CEILING FAN

Watch a ceiling fan from across the room: the **fan's slow swing** across the room is the trend, the **blades' rhythm** with each rotation is the seasonal component, the **wobble** from a loose screw is the cyclical murmur, and a **sneeze** that shakes the pole once is the irregular shock. Smoothing methods are you half-closing your eyes: the moving average blurs the blades into a steady glow, semi-averages take one glance at the left half and one at the right, and least squares draws the single straight swing that best explains the blur. Same room, same fan — the methods are just different squints.

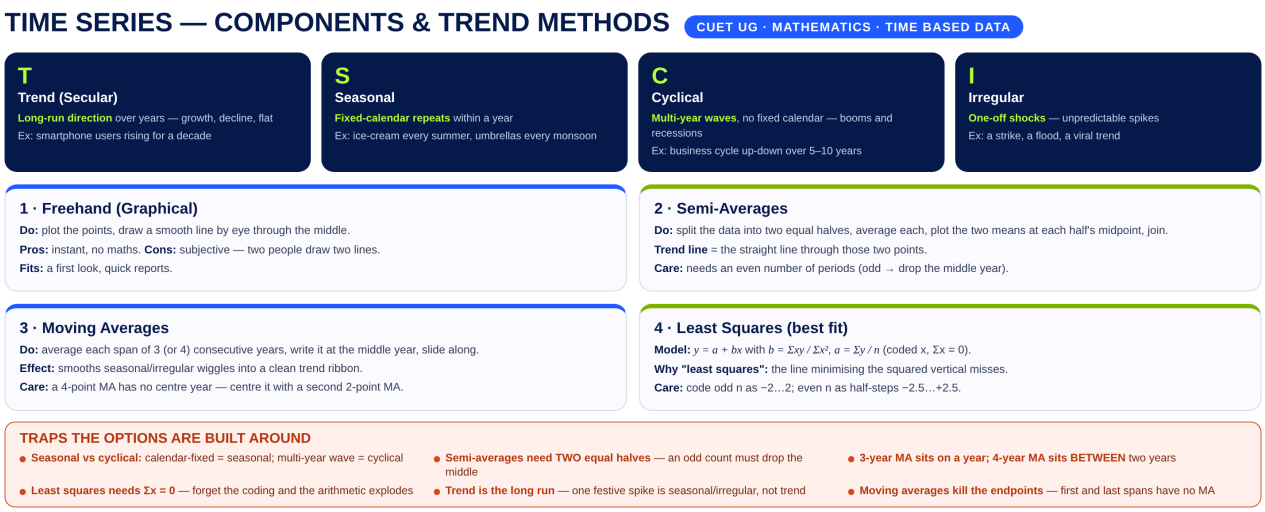


Figure 1: The unit on one board: the four components, the four measuring methods with their care points, and the traps between them.

Formula Cheatsheet — the Whole Chapter on One Page

FORMULA CHEATSHEET — TIME BASED DATA

Formula / rule	Letters mean	Fire it when
Time series = one variable + time stamps, in order	order carries meaning	identification
Additive model: $Y = T + S + C + I$	absolute swings	component questions
Multiplicative: $Y = T \cdot S \cdot C \cdot I$	swings grow with level	component questions
Seasonal: calendar-fixed, within a year	every year, same months	classification
Cyclical: multi-year wave, no calendar	booms and slumps	classification
Irregular: one-off shocks	strikes, floods	classification
Semi-averages: two EQUAL halves, two means, join	odd count drops the middle year	trend line

Formula / rule	Letters mean	Fire it when
Moving average: average each 3-year span, park at middle	slide the window	smoothing
4-year MA lands between years — centre it again	second 2-point pass	even-span MAs
Least squares: $a = \frac{\Sigma y}{n}$, $b = \frac{\Sigma xy}{\Sigma x^2}$	coded x, $\Sigma x = 0$	fitted trend
Odd n coding: $-2, -1, 0, 1, 2$	even: half-steps ± 2.5	least squares setup
Forecast: substitute the new x into $y = a + bx$	one substitution	prediction stems

Common Errors and Special Cases

COMMON ERRORS AND SPECIAL CASES — TIME SERIES

The mistake	What is right	The trap it sets
Calling any repeat "seasonal"	Calendar-fixed within a year = seasonal; multi-year = cyclical	Cyclical movements in seasonal clothing
Reading a spike as trend	Trend is the long run; spikes are S or I	Festive-month "growth" options
Semi-averages on an odd count directly	Drop the middle year first	Averaged-unequal-halves answers
4-year MA parked on a year	It lands BETWEEN two years — centre it	Misaligned MA tables
MA table with values at every year	The ends carry no MA	Full-column MA options
Least squares without coding x	Force $\Sigma x = 0$ by coding	Arithmetic explosions and wrong a
Mixing up a and b's meaning	a = value at central year; b = yearly change	Swapped-interpretation options
Forecasting by extending the raw series	Substitute into the FITTED line	Last-value-carried-forward lures

Sixty-Second Self-Test

Statement	Verdict
Shuffling a time series changes its meaning.	True — order is the data
A 5-year business boom is seasonal.	False — multi-year and calendar-free: cyclical
Semi-averages need an even number of periods.	True — two equal halves
A 3-year moving average assigns a value to every year.	False — the ends stay empty
With coded years, a equals the mean of y.	True — $\Sigma\frac{y}{n}$
The multiplicative model adds its components.	False — it multiplies them

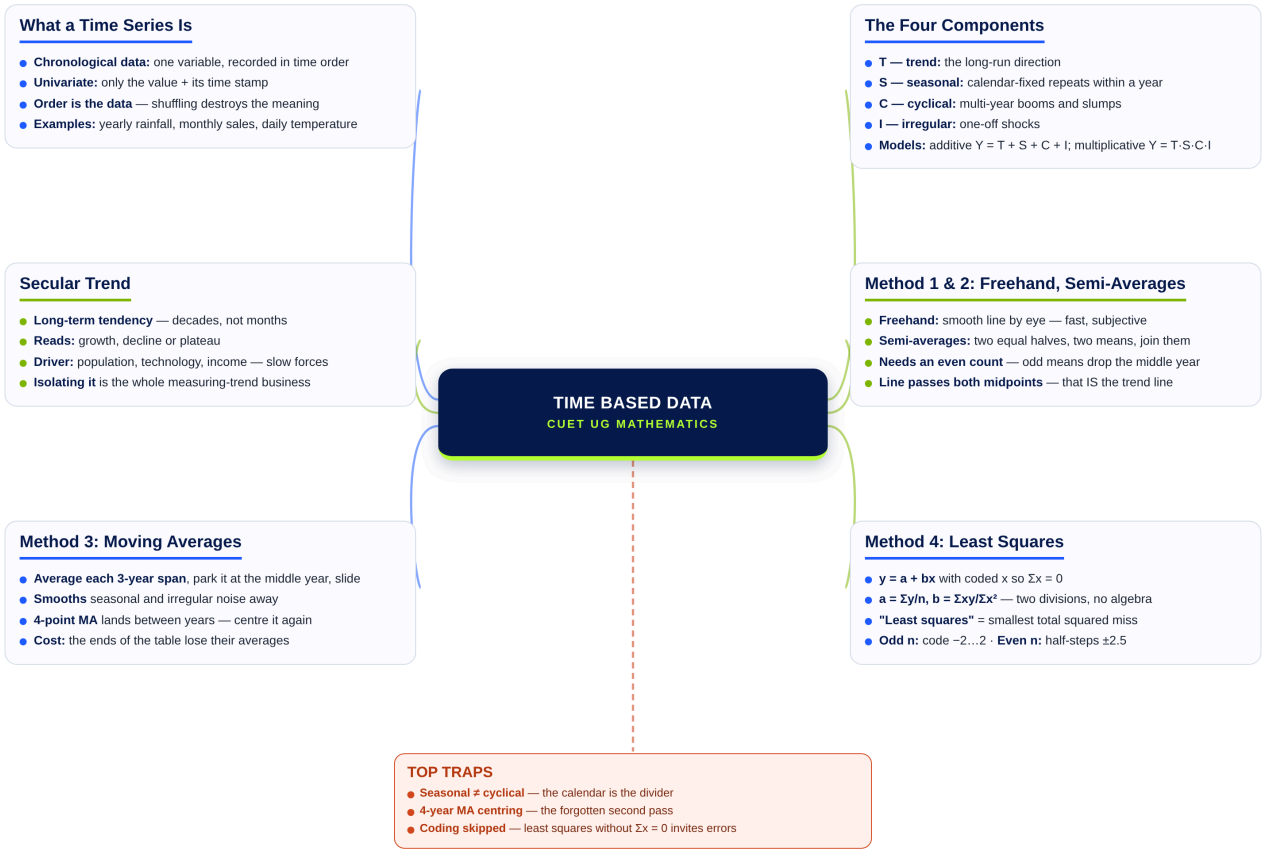


Figure 2: The chapter in one look: definition, four components, secular trend, four methods — and the traps under each.

Practice Questions — CUET UG Pattern (10 MCQs)

Ten questions in the CUET UG style. Each correct answer earns +5; each incorrect one costs −1. Classify first, compute second — most of these are verdict questions in disguise.

Q1. Which of the following is a time series?

(A) Marks of 50 students in one exam **(B)** Heights of 10 friends
(C) India's annual rainfall from 2015 to 2024 **(D)** Blood groups of 30 patients

Q2. Ice-cream sales peaking every May-June is an example of:

(A) seasonal component **(B)** trend component
(C) cyclical component **(D)** irregular component

Q3. A boom in the construction sector lasting about six years is best described as:

(A) seasonal **(B)** irregular
(C) cyclical **(D)** secular decline

Q4. Sales dropping for one week because of a warehouse fire reflects the:

(A) trend component **(B)** seasonal component
(C) cyclical component **(D)** irregular component

Q5. To apply the semi-averages method to 7 years of data, you should:

(A) average all 7 together **(B)** drop the middle year and split the rest equally
(C) split into halves of 4 and 3 **(D)** use moving averages instead

Q6. The 3-year moving average placed at the middle year for the values 12, 15, 14 is:

(A) 12.33 **(B)** 14.5
(C) 13.67 **(D)** 41

Q7. In the least squares trend line $y = a + bx$ with coded years ($\Sigma x = 0$), the slope b equals:

(A) $\frac{\Sigma xy}{\Sigma x^2}$ **(B)** $\frac{\Sigma y}{n}$
(C) $\frac{n}{\Sigma xy}$ **(D)** $\frac{\Sigma x}{\Sigma y}$

Q8. In the same coded setup, the constant a equals:

(A) the first year's value **(B)** the last year's value
(C) the slope of the line **(D)** $\frac{\Sigma y}{n}$ — the mean of the series

Q9. The model $Y = T \cdot S \cdot C \cdot I$ is called the:

(A) additive model **(B)** multiplicative model
(C) moving average model **(D)** least squares model

OurMocks

Q10. A 4-year moving average differs from a 3-year one because it:

(A) needs no calculation

(B) lands between two years and must be centred

(C) applies only to seasonal data

(D) cannot smooth a series

Detailed Solutions

Sol. Q1. A time series is one variable stamped with successive times — rainfall year by year qualifies. The others are cross-section snapshots: many individuals, one moment, no clock.

Answer: (C)

Sol. Q2. The peak repeats on a fixed calendar month every year — the signature of the seasonal component. A cycle would drift; a trend would not repeat.

Answer: (A)

Sol. Q3. Roughly six years, no fixed calendar anchor — a multi-year wave, hence cyclical. "Secular decline" contradicts the word boom.

Answer: (C)

Sol. Q4. A one-off shock with no pattern and no repeat date — irregular. The series did not wait for a season or a cycle; a fire happened.

Answer: (D)

Sol. Q5. Semi-averages demand two EQUAL halves; with 7 years the standard fix is dropping the middle year and averaging 3 + 3. Options (A) and (C) break the equality that the method runs on.

Answer: (B)

Sol. Q6. $\frac{12+15+14}{3} = \frac{41}{3} \approx 13.67$, parked at the middle year of the span. Option (D) is the sum, not the average — the un-divided slip.

Answer: (C)

Sol. Q7. With $\Sigma x = 0$ the normal equations collapse to $b = \frac{\Sigma xy}{\Sigma x^2}$ — one division over the coded products. The coding is what makes the formula this small.

Answer: (A)

Sol. Q8. The other collapsed equation: $a = \frac{\Sigma y}{n}$, which is exactly the mean of the series — the trend value at the central coded year $x = 0$.

Answer: (D)

Sol. Q9. Dots between components, not plus signs — the multiplicative model, used when the swings scale with the level of the series. The additive model uses $Y = T + S + C + I$.

Answer: (B)

Sol. Q10. An even span has no middle year, so the average lands between two years; the standard fix is a second 2-point centring pass. The smoothing power is unchanged — the placement is the difference.

Answer: (B)

Wrong-Answer Clinic

If you picked this	The thinking slip	The lasting fix
(A) in Q1	Matched "data" without checking the clock	Look for ONE variable across SUCCESSIVE times
(C) in Q2	Saw "big yearly business" as cyclical	Calendar-fixed = seasonal; the calendar is the divider
(A) in Q3	Any long movement labelled trend	Trend spans decades; 5-6 year waves are cyclical
(A) in Q5	Averaged everything for simplicity	Two EQUAL halves — the method's non-negotiable
(D) in Q6	Forgot to divide by 3	MA is an average — the sum alone is never the answer
(A) in Q9	Assumed components always add	Check whether swings grow with the level — that chooses the model

© OurMocks · www.ourmocks.com · CUET UG Mathematics · Time Based Data · 2027 cycle